INCH-POUND

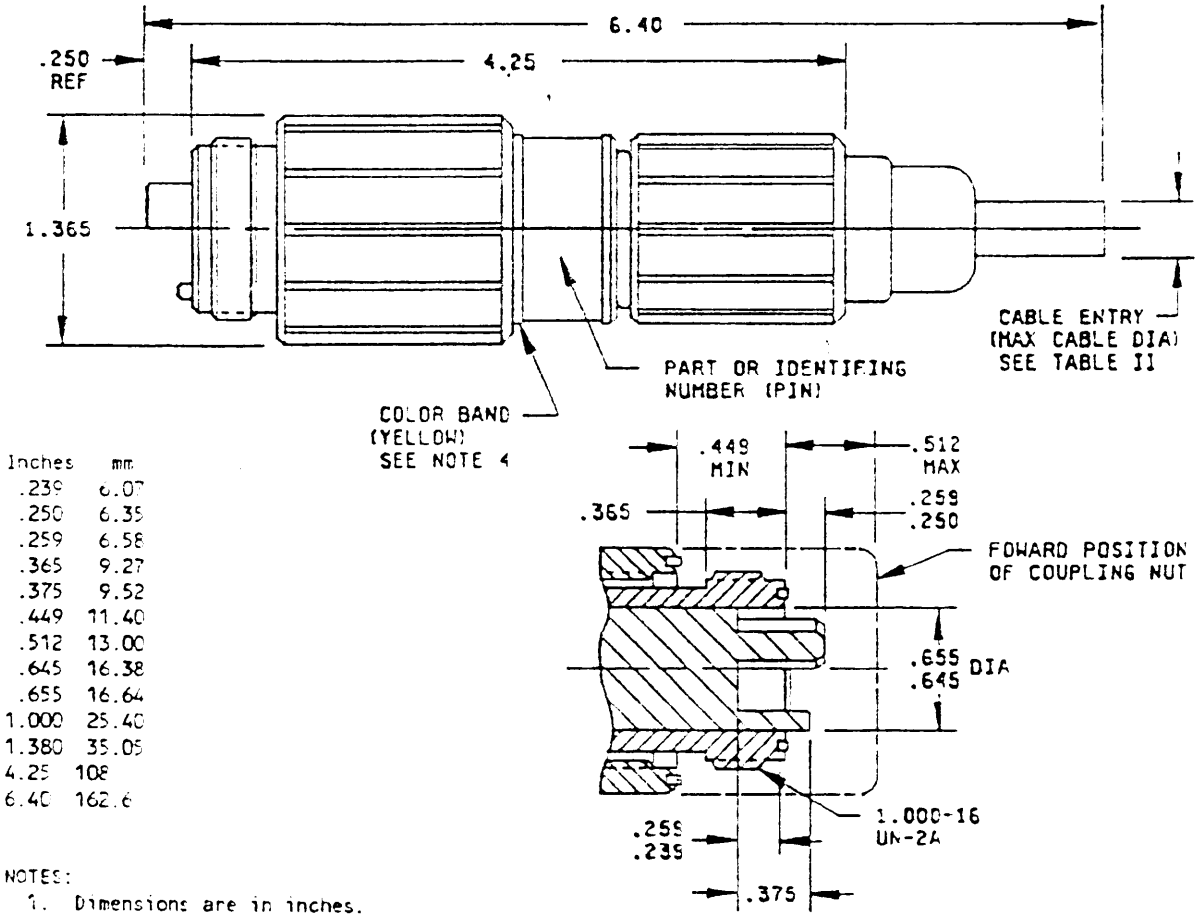
MIL-C-83526/1B
14 Oct 1993
SUPERSEDING
MIL-C-83526/1A
30 June 1989

MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, FIBER OPTIC, CIRCULAR, HERMAPHRODITIC, IN-LINE MOUNTING, 2 AND 4 POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83526.



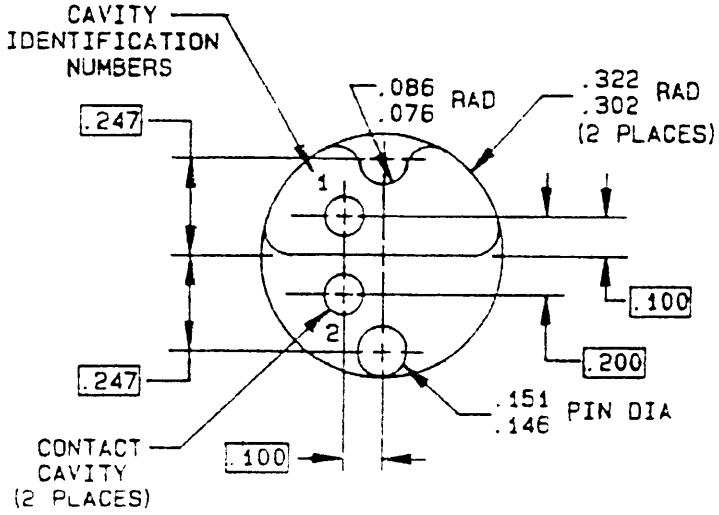
Inches	mm
.239	6.07
.250	6.35
.259	6.58
.365	9.27
.375	9.52
.449	11.40
.512	13.00
.645	16.38
.655	16.64
1.000	25.40
1.380	35.05
4.25	108
6.40	162.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± 0.010 (0.25 mm) for three place decimals and ± 0.03 (0.8 mm) for two place decimals. Angular tolerances are ± 0° 30'.
4. The phrase "Fiber-Optic" must be included within the yellow band.
5. Unless otherwise specified, dimensions are symmetrical about centerlines.

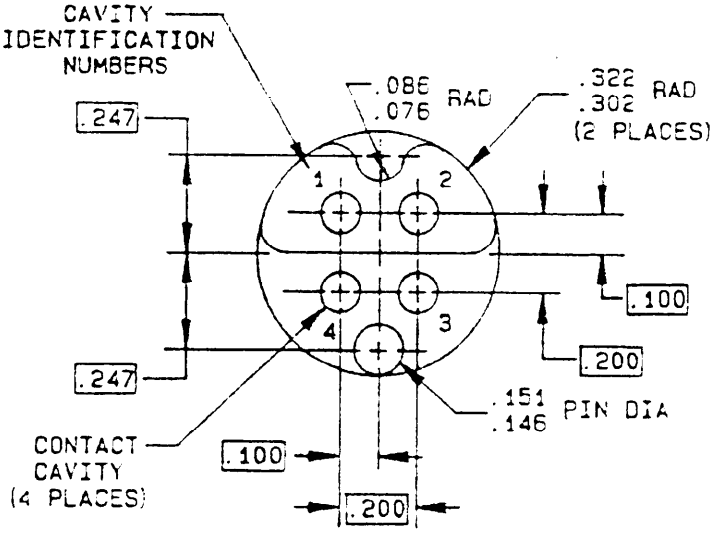
FIGURE 1. In-line connector.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.



2 Terminal connector type 1-01



4 Terminal connector type 1-02

Inches	mm
.076	1.93
.086	2.18
.100	2.54
.146	3.71
.151	3.84
.200	5.08
.247	6.27
.302	7.67
.322	8.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Terminal configuration face view.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2.

Fiber optic cable:

Cable diameter: In accordance with MIL-C-85045.

Cable configuration: In accordance with MIL-C-85045.

Fiber diameter: In accordance with MIL-F-49291.

Fiber numerical aperture: In accordance with MIL-F-49291.

Fiber type: Multimode.

Fiber class: Graded index.

Fiber composition: Glass/glass.

Materials: The material requirements are listed in table I.

Connector termination and assembly tools: As specified in MIL-K-83533.

Cleaning procedures: The cleaning procedures for the connector shall be included with the assembly instructions of the connector.

Termini: The required number of termini shall be supplied with each connector unless otherwise specified. Style P(pin) or S(socket) MIL-T-29504/1 and MIL-T-29504/2. For dummy terminus, see MIL-T-29504/3.

Weight: 8.0 ounces maximum.

Coupling forces: 11.0 pounds.

Coupling torques: 5 inch-pounds minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.

TABLE I. Material description.

Parts description	Material	Finish
Spacer Connector shell Insert (2 termini) Insert (4 termini) Compression nut Coupling nut	Aluminum alloy 2011-T3 QQ-A-225/3 " "	Anodized MIL-A-8625, type II class 2, color: gray
Strain relief Compression sleeve Strain relief sleeve	" " "	Chemical film MIL-C-5541, class 1A
Dowel pins Securing screw	Cres type 303 or 303SE per ASTM-A581/A582, condition A	Passivated MIL-S-5002
Strain relief boot O-rings	Fluorosilicone rubber MIL-R-25988, type I, class 1	None None
Shrink tubing	Melt seal polyolefin MIL-I-23053/4	None
S/H cap screw	MS 16995-9	None

PIN:

M83526/01 X X Y X

Basic specification sheet

Number of terminus positions required

Quantity	Designator
2	A
4	B

Cable size 1/

Cable OD in inches	Designator
0.178 - 0.216	1
0.217 - 0.256	2

Terminus supplied for actual fiber size 2/

Fiber OD in microns	Designator
123.0 - 124.4	A
124.5 - 127.0	B
138.0 - 139.6	C
139.7 - 142.0	D
No termini supplied	E

Keying position for terminus insert arrangement

Position	Designator
No alternate keying position required	3

1/ Select cable size and type in accordance with MIL-C-85045.
2/ Select fiber sizes and tolerances in accordance with MIL-F-49291.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.

Quality conformance and periodic inspection: The following cables shall be used for quality conformance and periodic inspection.

Number channels	Cables
2	PIN M85045/2B2A, 50 micron core, 125 clad, or equal.
4	PIN M85045/2C4A, 100 micron core, 140 clad, or equal.

For quality conformance and periodic inspection, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 3,994,564	11/30/1993
US 4,054,366	10/18/1994
US 4,060,308	11/29/1994

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - MC, OS, YD
Air Force - 13

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0100-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.